Lecture 8

Last time: Phasors, Single-phase, three-phase, power.

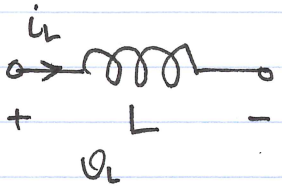
Today: Magnetics (Foundation of our electrical equipment – motors, generators, transformers.)

Q. Why in inductor?

$V = L\frac{di}{dt}$: We will "tear apart" L.

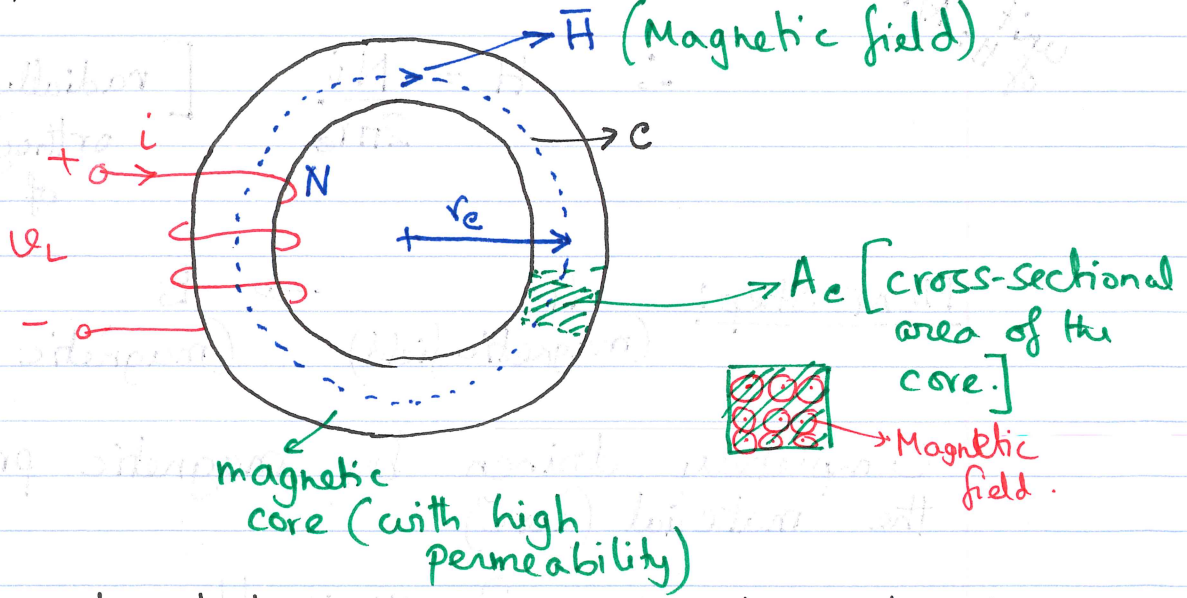
Q. Methods to analyse magnetic "circuits".

A.



$v_L = L\frac{di}{dt}$.

Let us play with Maxwell's equations:



We are interested in finding the relationship between voltage and current.

If we send a current 'i' through 'N' turns.

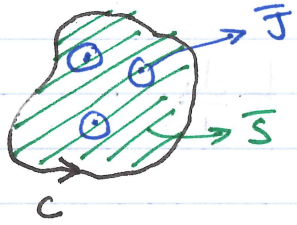
a magnetic field is established that "drives" magnetic flux in the magnetic core.

$$i \longrightarrow H \quad ?$$

Take help of Ampere's Law:

$$\oint_C \bar{H} \cdot d\bar{l} = \iint_S \bar{J} \cdot d\bar{A} + \iint_S \frac{\partial \bar{D}}{\partial t} \cdot d\bar{A}$$

Current flow creates magnetic field.



Magnetoquaristatic Problem.
i.e wavelength $\gg$ device dimensions.

$$H \cdot 2\pi r_e = Ni$$

unit of H: A/m.

$$\therefore \quad H = \frac{Ni}{2\pi r_e}$$

[radially directed along the contour orthogonal to area of crosections of core].

Next Step: H (magnetic field) $\longrightarrow$ B (magnetic flux density).

Completely driven by magnetic property of the material (core).

$$B = \mu_0 \mu_r H$$

permeability of free space: $4\pi \times 10^{-7}$ H/m (Henry/m).

relative permeability of the core material: unitless.

$$\therefore \quad B = \left[\frac{\mu_0 \mu_r}{2\pi r_e}\right][Ni] \quad \text{unit : Tesla (T).}$$

Since, flux density : flux flowing / unit area.

$$\therefore \quad \underset{\text{magnetic flux}}{\phi} = \left[\mu_0 \mu_r \cdot \frac{A_c}{2\pi r_e}\right] Ni$$

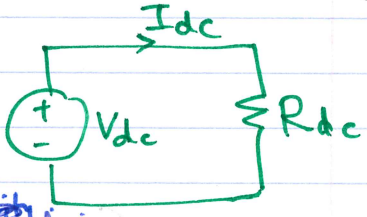
$$= \frac{Ni}{\left(\frac{l_e}{\mu_0 \mu_r A_c}\right)}$$

where $l_e = 2\pi r_e$ ← mean path length.

This expression very similar to :

$$I_{dc} = \frac{V_{dc}}{R_{dc}} = \left(\frac{l}{\rho A}\right)$$

resistivity ~~conductivity~~.



As if, we can build an analogy :

Electric Ckt :	Magnetic circuit.
Voltage (V_{dc})	Magnetomotive force (MMf) Ni
Current (I_{dc})	Flux (ϕ)
Resistance (R_{dc})	Reluctance ($\mathcal{R}$) $= \frac{l_c}{\mu_0 \mu_r A_c}$.

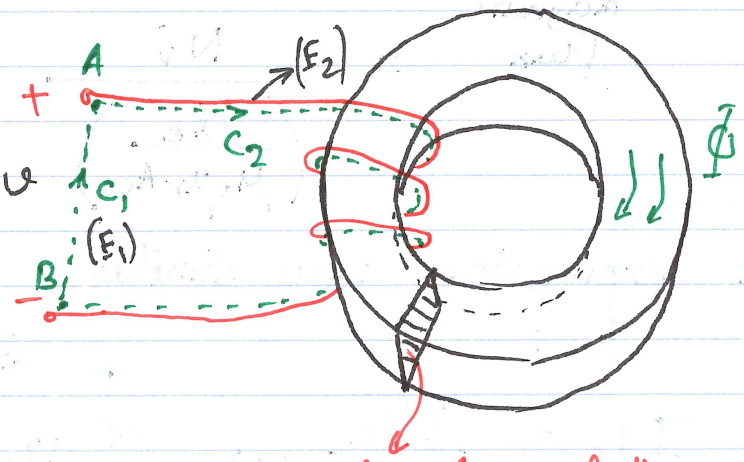
This equivalence will "simplify" our life in analysing magnetic equivalent circuits".

Next Step : $B \longrightarrow v$?

Take help of Faraday's Law :

Changing magnetic field creates voltage

$$\oint_C \bar{E} \cdot d\bar{l} = -\frac{d}{dt} \iint_S \bar{B} \cdot d\bar{A} \quad .$$



A_c : Area of the core.

$$\int_{C_1} \bar{E}_1 \cdot d\bar{l} + \int_{C_2} \bar{E}_2 \cdot d\bar{l} = -\frac{d}{dt}[N\phi]$$

or, $-v + 0 = -\frac{d}{dt}\lambda$ ← Flux linkage. ↓ flux times number of turns.

assuming infinite conductivity of the wire.

$$\therefore \quad v = \frac{d\lambda}{dt} = N\frac{d}{dt}\phi .$$

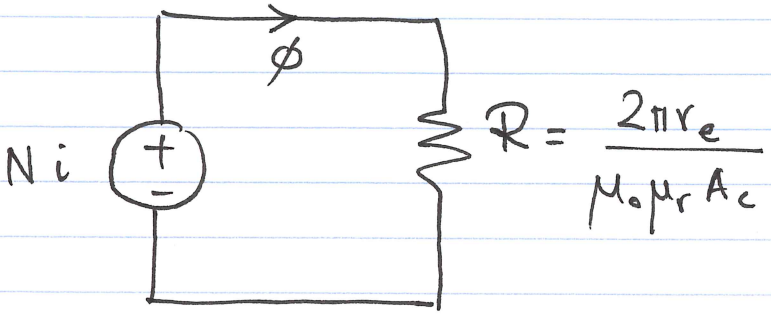
or, $$v = N \frac{d}{dt} \cdot \left[\frac{1}{R}\right] Ni$$

$$= \frac{N^2}{R} \frac{d}{dt} i$$

$$L = \frac{N^2}{\mathcal{R}}$$

N² ← no. of turns

$\mathcal{R}$ ← Reluctance.

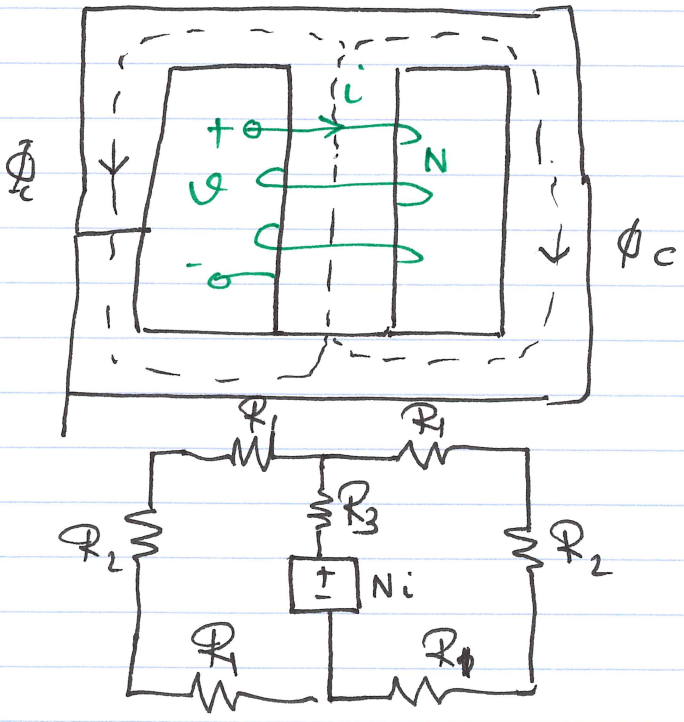


Magnetic equivalent circuit

Reluctance network model.

$L = f$ (No. of turns, geometry, material property).

Q. How do we figure out inductance



Calculate equivalent $\mathcal{R}$.

$\therefore L = \frac{N^2}{\mathcal{R}}$.

